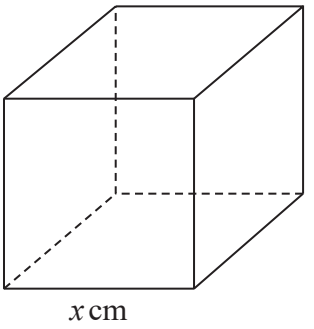
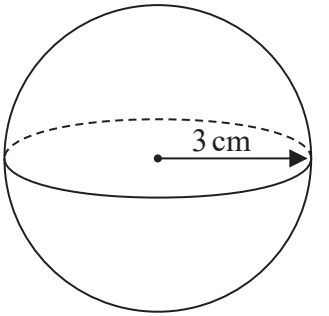
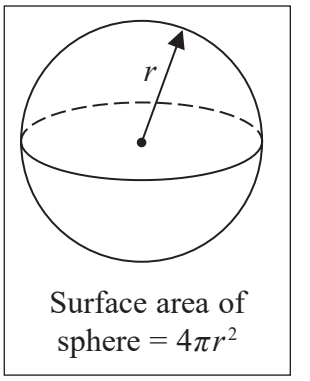
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**7** The diagram shows a cube with edges of length $x$ cm and a sphere of radius 3 cm.




The surface area of the cube is equal to the surface area of the sphere.

Show that $x = \sqrt{k\pi}$ where $k$ is an integer.

Cube: 6 faces, each area $x^2$, so area = $6x^2$

Sphere: $4\pi r^2 = 4\pi(3^2) = 36\pi$

Equal areas: $6x^2 = 36\pi$

Divide by 6: $x^2 = 6\pi$

$x > 0$, therefore $x = \sqrt{(6\pi)}$, so $k = 6$.

**1 mark**
cube area $6x^2$

**1 mark**
sphere area $36\pi$

**1 mark**
equate the two areas

**1 mark**
$x = \sqrt{(6\pi)}$; $k = 6$

**(Total for Question 7 is 4 marks)**